

Agent-swarm physics: hypothesis compendium

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(Living status document, regenerated 2026-10-04 12:08. Not the paper.)

Two RevTeX pages per hypothesis follow; click a row to jump to it. Ratings are estimates calibrated across hypotheses by the coordinator (rubric: [writeup/hypothesis-pages/RUBRIC.md](#)). **Completion:** estimated progress of the research direction (idea 5%, round 1 done 35–50%, holdout run 55–65%, robustness and causal designs 70–85%, settled 90–100%). 5% 25% 50% 75% 100% **Credence** (scoring v2, [writeup/scoring/scoring-v2.pdf](#)): probability that the scored claim (gray line under each title) survives an independent confirmatory test at its stated scope, built from an evidence ladder; “!” marks a fragile claim (its headline changed between data versions). Red = low, gray = 0.5, blue = high. **EU** (expected usefulness) = credence $\times$ value if true (0–5: whose decision changes, by how much, how generally); gray $\approx$ 1.5. Six anchors were double-scored by a blind rater (mean $|\Delta p| = 0.11$). Credence 0 0.2 0.4 0.5 0.6 0.8 1

ID	Hypothesis (scored claim)	Complete	Credence	EU	p.
H01	Emergent superagents exist No coordination-defined group is an effective (KW) superagent; the persistent unit is agent + own artifact	45%	0.61	1.5	6
H02	Inferred couplings reflect real influence Activity-timing pairwise couplings sit at the null floor; regime-III ‘collective coupling’ was day edges	65%	0.95!	2.4	8
H03	Swarm activity is self-exciting, and its criticality is set by the go... Activity is subcritical everywhere ($n \approx 0.2$ – 0.4)	50%	0.89	1.3	10
H04	External forcing reshapes the response kernel, reversibly A first nudge adds ~1.2 active minutes per 30 (A30 1.16 [0.75, 1.57]) to its named target only, starting at the recei...	65%	0.48!	0.7	12
H05	Cutting the cross-room channel lowers entropy production, and rooms b... Rooms are coupled blocks for talk, and the coupling runs through reading: co-location adds nothing once ledger read c...	55%	0.86	2.6	14
H06	Free weeks show neutral cooperative dynamics with a cooperator core Agents state private sub-tasks everywhere; work concentrates on a shared repo when one exists (neutral copying at bes...	50%	0.84	1.7	16
H07	The RPG forks diverge from a common ancestor at a measurable rate Isolated copies diverge gradually and independently; identical new content arises only by convergent repair with no r...	50%	0.84	1.7	18
H08	Context is the coupling Responses are gated at the recipient’s next model call (read-out)	50%	0.93	2.8	20
H09	Agent swarms have an effective thermodynamics Agents leave idle at a timer gate; unaddressed chatter holds them there (OR 0.50) and a mention releases them (OR 3.1...	40%	0.47!	0.9	22
H10	Goals are Legendre pushes: a swarm’s response to an assigned goal is ... Goal kickoffs push content along the goal direction, but pre-kickoff fluctuations do not predict who moves (no linear...	50%	0.82	1.6	24
H11	Division of labor vs herding is the sign of a Potts coupling, set by ... Agents herd onto shared projects in attention and work; ownership disperses	50%	0.84	2.1	26
H12	Groupthink is dimensional collapse: few collective modes, shrinking p... Talk and content collective modes are real; the activity market mode was an artifact; low-PR weeks are restatement lo...	52%	0.81!	1.6	28

ID	Hypothesis (scored claim)	Complete	Credence	EU	p.
H13	Model families carry their own fields, and couple by family Model-family content fields are writing style; behavior carries only a weak family trace (tooling-like)	50%	0.92	1.9	30
H14	Entropy production of behavior-state sequences: per-family arrows of ... Fine-action irreversibility is real for every agent and survives scaffold removal; the regime contrast was scaffold	50%	0.72!	0.7	32
H15	Semantic information through natural scrambles: which information kee... A forced wipe cuts work commits ~39% for ~10 turns ($\approx 10\%$ of a segment's output); saving more memory doesn't soften it...	55%	0.86!	2.6	34
H16	Agents get stuck in metastable traps and escape by Kramers kinetics; ... Traps age (power only in #51); one directed kick at a pause gate helps; error-loop aging was a stderr artifact	48%	0.44!	1.1	36
H17	Behavior is a Markov state model with a few metastable sets; its mixi... Behavior is non-Markov; the slow split is acting vs waiting; no metastable stuck states	45%	0.57	0.6	38
H18	Attention dilutes as $1/k$: response to a message falls with the number... Per-pair attention dilutes as $k^{-0.66}$ in every period	48%	0.89	2.2	40
H19	One curve for all periods: per-period loop gains collapse onto a func... No common loop-gain curve; trimmed activity gains track Hawkes gains; the channel split was day edges	50%	0.33!	0.3	42
H20	Aging: day-to-day content autocorrelation depends on time since kicko... Content does not age: it settles within days, then decorrelates at an age-independent rate	50%	0.48	0.7	44
H21	The debate week (#12) is a two-sublattice antiferromagnet Assigned debate sides show two-sublattice order in stance (replies), not in content; assignment alone explains it	50%	0.86	1.3	46
H22	Private, conflicting goals make #51 a spin glass #51 private goals produce no frustration: rivals' content is more similar (homophily) and stance is ferromagnetic	50%	0.68	1.0	48
H23	Distilling the village into a fine-tuned leader copies its vocabulary... A distilled leader's messages are no more corpus-like than base Kimi's (a base-model echo), and none of its plans car...	40%	0.57!	0.9	50
H24	In the forecast week (#21), switching from independent drafts to comp... In the forecast week, reading a teammate's document moves an agent's statements toward that teammate; there is no agg...	50%	0.35!	0.7	52

ID	Hypothesis (scored claim)	Complete	Credence	EU	p.
H25	A distance-to-criticality dial, T/T _c , per window and per channel Daily equal-time gains are subcritical (~ 0.1 – 0.16); content 'near-critical' is a shared field	50%	0.84	1.3	54
H26	The swarm is near-critical in what it says but subcritical in when it... Content and talk co-move within rooms at a loop gain of ~ 0.5 (not near-critical); activity has no room gain once the ...	50%	0.89!	1.8	56
H27	Critical slowing down warns of herding waves hours ahead Critical slowing down does not warn of herding onsets	50%	0.36	0.5	58
H28	Links are the contagion vector of herding Links co-move with project switches in attention and work ($HR \approx 2.6$) but mark bursts rather than cause them: recipien...	52%	0.80	1.6	60
H29	Driver nodes: where an operator message moves the whole swarm Content influence is address-gated (named 0.05–0.08 per message vs unnamed ≈ 0 in #51); content-pull driver rankings ...	50%	0.65	1.9	62
H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm tod... A first nudge adds ~ 1 active minute to its target ($\chi_{\text{act}} \approx 1.0$ in G51 and G38) and works in swarm-wide lulls; human c...	50%	0.64!	1.3	64
H31	Consensus time scales with the interaction graph's spectral gap Consensus time doesn't follow $1/\lambda_2$ (constant ~ 4 h forecast); deadline votes settle at read-out speed	50%	0.65	1.6	66
H32	A net information current identifies de facto leaders Exposure-gated content transfer is real (17/32 periods; read beats unread at 60 s in 17/17), but about a third of it ...	50%	0.87	1.7	68
H33	The diversity–productivity curve is an inverted U: the swarm's operat... No diversity–productivity optimum on committed work; the #51 tail is model-dependent	50%	0.57	0.8	70
H34	Idea cascades follow a power law whose exponent reads off the distanc... Idea cascades are subcritical (R 0.09–0.40) and exposure-locked in 32/32 periods, but about half of the lock is conv...	50%	0.89	1.8	72
H35	The nudger is a measurably inefficient Maxwell demon Nudges buy glances (+0.08 P(active within 30 min) per first nudge), not committed work; switching the nudger on or of...	55%	0.63!	0.9	74
H36	A reorganization alarm: susceptibility and multi-information peak at ... Plain topic shift detects goal changes (AUC ~ 0.95) far better than the physics alarm (AUC 0.66–0.68, not better than ...	50%	0.86!	2.1	76

ID	Hypothesis (scored claim)	Complete	Credence	EU	p.
H37	Conflict lives in stance, not topic: stance spins are antiferromagnetic Stance recovers assigned debate teams; no stance conflict without assignment	40%	0.71	1.8	78
H38	Joint silences are platform stalls Regime-III co-activation is 70–80% the runner’s start/stop schedule	50%	0.94!	2.8	80
H39	Catalysts vs. fields No lever is a pure catalyst: nudges both unstick (escape $\times 1.3$ – 1.6) and tilt (idle/wait share -5 to -8 points); mentio...	50%	0.74	1.9	82
H40	The call clock sets the coupling In the computer-use scaffold (regimes II–III), reply coupling runs on the call clock ($\eta \approx 0$): per-hour coupling = per...	45%	0.65	1.9	84
H41	Read-out gating gives the swarm a light cone Inside rooms, novel items follow the logged light cone (acausal $\leq 4\%$, median 0.6%) and rooms cage spread (NE42 moved ...	45%	0.69	1.7	86
H42	Cross-excitation is a delayed step at the next read-out Talk timing follows each agent’s call clock; cross-triggering ≈ 0 except named messages	42%	0.66	1.3	88
H43	Kicks leave a refractory window No refractory window beyond one read-out; only same-call kicks are wasted	40%	0.67	2.0	90
H44	Erasure makes agents busy but unproductive Forced erasure triggers orderly re-reading, costs 4–11% of output, breaks loops	42%	0.91	3.2	92
H45	Context homeostasis: each agent keeps its context near a set point Room share of context is passive; the scaffold bounds context	40%	0.80	1.6	94
H46	Style is a conserved charge Style drifts with context fill and resets at erasure, yet identifies agents better than content	42%	0.80	2.4	96
H47	Rooms set the coherence length Rooms bound content coherence when they do different things; inside rooms coherence follows conversation	40%	0.68	1.7	98
H48	The settling time is a mixing time on the read-out graph Coverage takes minutes, settling hours; the scaling test is unpowered	40%	0.18	0.2	100

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H49	Edge-trimmed regime III is a dilute ferromagnet Edge-trimmed regime-III excess is a dense/shared-field mode, not a dilute ferromagnet	40%	0.80	1.2	102
H50	Field or coupling: transfer functions and lag tell which Activity co-movement is the scheduler’s field; talk is a coupling gated at hop 1	45%	0.86	2.6	104
H51	One dial: an effective coupling collapses the phase diagram (draft) specified (promoted 2026-10-04 from HH178; round 1 waits for its data-quality inputs)	–	–	–	106
H52	Humans are just loud agents Humans get a status-weighted attention premium ($\sim 2\times$ replies), not deference	45%	0.64	1.9	108
H53	Herdin is announcement-seeded nucleation Herdin is seeded by a project’s first chat link; deadline votes are read-out-bound	42%	0.56	1.7	110
H54	The kickoff text is the quench target Day-1 content lands on what the kickoff names; agents land on private goals	45%	0.94	2.6	112
H55	Norm-enforcers are the swarm’s immune cells Correctors get warmer replies; corrections rarely reach looping agents	40%	0.46	0.7	114
H56	Entropy production fingerprints the platform Turn-level EP does not detect platform changes; log features do	40%	0.67	1.3	116
H57	Copying beats transformation when the backlog is large Backlog doesn’t raise echo; near-duplication is mostly contemporaneous convergence	40%	0.56	0.8	118
H58	Effective superagents are coordinated agents plus their artifacts Where there is power (#51b–d), no coordination-defined unit is an effective superagent: co-movement onto artifacts is...	40%	0.76	1.9	120
H59	One lever model for all operator inputs No single (field, catalysis, read-out delay) triple transfers across operator input classes (leave-one-class-out fail...	40%	0.63	1.2	122
H60	An index policy beats the nudger (draft) specified (promoted 2026-10-04 from HH252; round 1 can start now)	–	–	–	124

ID	Hypothesis (scored claim)	Complete	Credence	EU	p.
H61	Contagiousness is predictable at first use (draft) specified (promoted 2026-10-04 from HH253; round 1 can start now)	—	—	—	126
H62	Ideas travel the reply graph, not the room (draft) specified (promoted 2026-10-04 from HH254; round 1 can start now)	—	—	—	128
H63	Herding bursts start with a work signal (draft) specified (promoted 2026-10-04 from HH263; round 1 can start now)	—	—	—	130
H64	Conflict needs a scarce prize (draft) specified (promoted 2026-10-04 from HH255; round 1 can start now)	—	—	—	132
H65	Leaders are routers, not sources (draft) specified (promoted 2026-10-04 from HH264; round 1 can start now)	—	—	—	134
H66	Shared platform latency is a common-noise field (draft) specified (promoted 2026-10-04 from HH256; round 1 can start now)	—	—	—	136
H67	A lagged criticality dial (draft) specified (promoted 2026-10-04 from HH262; round 1 waits for a running agent)	—	—	—	138
H68	Attention dilution is a mixture of two strategies (draft) specified (promoted 2026-10-04 from HH258; round 1 can start now)	—	—	—	140
H69	Restatement loops are context fixed points (draft) specified (promoted 2026-10-04 from HH259; round 1 can start now)	—	—	—	142
H70	The semantic information of the artifact store (draft) specified (promoted 2026-10-04 from HH261; round 1 can start now)	—	—	—	144
H71	Memory is a first-order homeostat (draft) specified (promoted 2026-10-04 from HH269; round 1 can start now)	—	—	—	146
H72	Trap aging is input starvation (draft) specified (promoted 2026-10-04 from HH260; round 1 can start now)	—	—	—	148
ID	Hypothesis (scored claim)	Complete	Credence	EU	p.
H73	Style = weights + context + register (draft) specified (promoted 2026-10-04 from HH265; round 1 can start now)	—	—	—	150
H74	An operator-grade change detector (draft) specified (promoted 2026-10-04 from HH268; round 1 waits for a running agent)	—	—	—	152
H75	A thermodynamic speed limit on re-allocation (draft) round 1 launching (card, observables and predictions written 2026-10-04 ~19:10 UTC, before any allocation statistic o...	—	—	—	154
H76	Excess vs housekeeping irreversibility (draft) round 1 launching (card, observables and predictions written 2026-10-04 ~19:20 UTC, before any flux or EP statistic o...	—	—	—	156
H77	Repos as replicators: the selection-resolution bound (draft) specified (promoted 2026-10-04 from HH325 by Vivian); round 1 card written 2026-10-04 before any outcome statistic on...	—	—	—	158
H78	Replicator growth order: herding vs division of labor (draft) specified (promoted 2026-10-04 from HH326 by Vivian); round 1 card written 2026-10-04 before any outcome statistic on...	—	—	—	160
H79	An artifact-only autocatalytic set (draft) specified (promoted 2026-10-04 from HH327 by Vivian; round 1 launching)	—	—	—	162
H80	Assembly index vs compression as an agent signature (draft) specified (promoted 2026-10-04 from HH328 by Vivian; round 1 launching)	—	—	—	164